I²C master design in Verilog

Laboratory 11 | Register-level temperature acquisition with the onboard ADT7420

Design objective

Implement the transaction controller for the ADT7420 on the Nexys4 DDR. Select the temperature register, change transfer direction with a repeated START, acquire two bytes and transmit a signed decimal temperature over UART. The implementation uses synchronous Verilog throughout.

Parameter	Specification
Bus interface	100 MHz FPGA clock, nominal 100 kHz I²C, single controller
Measurement	13-bit signed temperature, one acquisition per second
Host interface	115200 baud, 8N1, ASCII degrees Celsius followed by CR/LF

Verification objective

Identify the driver of SDA at each acknowledgement clock. Demonstrate recovery from a missing sensor ACK and verify both positive and negative records at the serial output pin.

Temperature conversion and address space

The ADT7420 performs conversions internally and maintains its latest result in two output registers.

Address	Register	Function
0x00	Temperature MSB	Holds the upper eight bits, including the sign.
0x01	Temperature LSB	Bits 7:3 complete the 13-bit code.
0x03	Configuration	Power-up default selects continuous conversion and 13-bit resolution.

Device selection and register selection

The seven-bit device address 0x4B selects the ADT7420 on the bus. The pointer byte 0x00 selects a location inside that device. During the sequential read the sensor increments its pointer, so the second byte comes from 0x01.

A normal conversion typically takes 240 ms. The FPGA waits one second after reset and then reads once per second. Host record spacing reflects the FPGA scheduler. The sensor's conversion clock runs independently.

Open-drain signaling and bus ownership

A pull-up restores the high level once every device connected to the line releases its driver.

```
assign TMP_SDA = sda_low ? 1'b0 : 1'bz;
```

When sda_low is zero, the FPGA disconnects its output driver. Its input buffer continues to observe the physical pin. During a write ACK the FPGA releases SDA and the sensor can pull it low. In simulation, tri1 models the pulled-up line shared by the two output drivers.

Bus event	SCL	SDA behavior
START or repeated START	High	Falling edge begins an address phase.
Data or acknowledgement	High	SDA remains stable while the receiver samples.
STOP	High	Rising edge releases the bus.

The supplied XDC maps TMP_SCL to C14 and TMP_SDA to C15 and enables pull-ups. Both pins are bidirectional. A synchronized SCL input lets the engine wait for the released clock to rise.

Combined register read

Each byte occupies eight data clocks followed by a ninth acknowledgement clock.

```
S   96 A   00 A   Sr   97 A   MSB A   LSB N   P
```

Field	Data driver	Driver and level on clock 9
0x96 / address + W	FPGA	Sensor drives low for ACK.
0x00 / register pointer	FPGA	Sensor drives low for ACK.
0x97 / address + R	FPGA	Sensor drives low for ACK.
Temperature MSB	Sensor	FPGA drives low to request the second byte.
Temperature LSB	Sensor	FPGA releases SDA for the final NACK.

Bytes 0x96 and 0x97 encode (0x4B << 1) with R/W in bit 0. Sr occurs while the transaction is still open. P ends the transaction. A target NACK during a write invalidates the acquisition.

SCL timing and byte transfer

The supplied engine executes START, WRITE, READ_ACK, READ_NACK and STOP primitives.

0: set SDA > **1: release SCL** > **2: sample pin** > **3: pull SCL low**

At 100 MHz, 250 system clocks give a quarter-period of 2.5 µs. Four phases produce a nominal bit period of 10 µs. Each byte primitive repeats them over slots 0 through 8. The engine extends the high interval for START and STOP and samples the synchronized input during phase 2.

Turnaround between read bytes

Interval	Sensor	FPGA
Bit 0 of MSB	Drives the final data bit	Releases SDA and samples the pin.
Ninth clock	Releases SDA	Asserts ACK by pulling SDA low.
Bit 7 of LSB	Drives the next data bit	Releases SDA before SCL rises.

Implement the transaction sequencer

Optional RTL exercise: complete the separate template in student/temperature_sequence.v.

State	Role
IDLE	Wait for the timer, initialize the step index and clear the failure flag.
ISSUE	Present one primitive and assert start for one system clock.
WAIT_DONE	Advance on done. Capture received or evaluate nack as appropriate.

1. Decode the eight steps on page 4. Select the primitive and the transmit byte for each WRITE. Use the START primitive for both S and Sr.

2. Capture the MSB and LSB on their respective done pulses. If the target NACKs a write, set failed and branch directly to STOP. Leave the previous committed temperature intact.

3. On completion of a successful STOP, assemble the signed temperature and pulse valid for one cycle. Return to IDLE. Explain why the formatter must receive only complete samples.

Signed reconstruction and decimal scaling

In the default sensor mode, word bits [15:3] represent temperature in units of 1/16 °C.

MSB, LSB	13-bit code	Signed count	Serial output
0C 80	0x0190	+400	T=+25.00 C
FA C0	0x1F58	8024 - 8192 = -168	T=-10.50 C
00 08	0x0001	+1	T=+00.06 C (rounded)

Fixed-point implementation

Concatenate the full MSB with bits [7:3] of the LSB into a signed 13-bit register. Save the sign and form an unsigned magnitude. To round to hundredths of a degree, compute floor((100 × magnitude + 8) / 16). This rounds half steps away from zero.

The formatter converts the scaled magnitude to decimal by shift-and-add-3. Every record contains a sign, at least two integer digits, two fractional digits, a space, C, carriage return and line feed. Sensor resolution is 0.0625 °C even though the terminal displays hundredths.

Behavioral verification in Vivado

Install Vivado 2022.2 with XSim and Artix-7 support. Simulation runs without the board or Python.

```
vivado -mode batch -source create_project.tcl
```

1. Run the command in the extracted Lab11 folder. Open build_reference/lab11.xpr. The project uses the complete transaction controller in reference/temperature_sequence.v.

2. Run Behavioral Simulation with tb_lab11. Add scl, sda, sensor_low, dut.sda_low, command, done, nack, temperature, valid and tx from their respective scopes. Restart and select Run All.

3. The target model withholds the first address ACK. Check that STOP follows and valid stays low. The next two transactions return the positive and negative test words. An independent UART monitor checks the decoded records, including framing and line endings.

Acceptance criteria

Five START edges, three STOP edges, two committed temperatures and the PASS message. The simulation reduces timer and divider values to shorten execution.

Implementation and bench acquisition

Target xc7a100tcsg324-1. Connect the powered Nexys4 DDR through a PROG/UART USB data cable.

Port	CLK100MHZ	CPU_RESETN	TMP_SCL	TMP_SDA	UART_RXD_OUT
Pin	E3	C12	C14	C15	D4

1. After simulation passes, generate the bitstream. Review the routed timing summary for the 10 ns clock. In Hardware Manager, open the target and program lab11_top.bit. The first request starts about one second after reset release.

2. Open the board's COM port in PuTTY at 115200 baud, 8 data bits, no parity, 1 stop bit, flow control off. Record ten lines. They measure local sensor temperature, which depends on board heating, thermal contact and airflow.

3. For an empty terminal, check the USB port and UART settings first. Inspect the ninth SCL pulse after 0x96 in simulation or with an ILA. A missing ACK deliberately suppresses output for that acquisition.

Experimental report and assessment

Submit a labeled waveform, a serial capture and an explanation of the controller transitions.

Time	Activity	Evidence
20 min	Interface and addressing	Identify the SDA driver on each ninth clock.
40 min	Transaction FSM	Trace the step decoder and NACK recovery path.
30 min	Simulation	Label Sr, first-byte ACK, final NACK and both UART lines.
30 min	Board measurements	Record ten temperatures and calculate their mean.

Explain why a high SDA level on the ninth clock can mean a failed write or a successfully completed read. Estimate the wire time for five bytes including acknowledgement clocks. Account for the difference between this time and the one-second reporting interval.

Analog Devices | ADT7420 Rev. A: conversion modes, temperature data format and serial interface

Digilent | Nexys4 DDR reference manual, section 13: temperature sensor and I²C address

Digilent | Nexys4 DDR Rev. C master XDC: temperature sensor, clock, reset and USB-UART